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Evaluating Logical Reasoning in Large Language Models Through Fine-Tuning and Neuro-Symbolic Bridging

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Anonymous submission

***Abstract*—Large language models can learn controlled deductive tasks, while neuro-symbolic pipelines seek to make their inferences executable and inspectable. Whether a symbolic bridge helps an already fine-tuned model depends on natural-language-to-logic translation fidelity. We present a matched evaluation on 200 held-out ProofWriter examples comparing a base Qwen3-1.7B model (C1), the same model with a checkpoint-250 LoRA adapter (C3), and that identical fine-tuned checkpoint routed through an NL2Logic-to-Z3 symbolic bridge whose verdict is the final answer (C4). Under strict scoring, with invalid outputs counted as errors, accuracy rises from 48.5% in C1 to 82.5% in C3, while macro-F1 rises from 0.251 to 0.826. Adding the bridge reduces accuracy to 49.5% and macro-F1 to 0.459. C4 corrects 12 C3 errors but overturns 78 correct C3 predictions; the -33.0 percentage-point change is significant by exact McNemar testing ($p = 5.20 \times 10^{-13}$). Although 89.5% of bridge outputs execute, 80 executable verdicts are wrong and translation traces report widespread semantic divergence. In this implementation, translation fidelity is the principal bottleneck.**

***Index Terms*—logical reasoning, large language models, fine-tuning, neuro-symbolic reasoning, semantic parsing, first-order logic, Z3**

I. Introduction

ProofWriter measures open-world deduction by pairing natural-language theories with conclusions labeled True, False, or Unknown [1]. Neuro-symbolic systems such as Logic-LM and LINC translate language into formal representations before solver-based inference [2], [3]; NL2Logic constrains this translation through a recursively constructed abstract syntax tree (AST) [4]. Their success therefore depends on preserving the semantics of the original problem.

We isolate fine-tuning and bridge effects on one frozen 200-example ProofWriter subset [1]. C1 is base direct inference, C3 is fine-tuned direct inference, and C4 routes the identical C3 checkpoint-250 adapter through the symbolic bridge, whose verdict becomes the final prediction. Accordingly, C1→C3 measures fine-tuning, C3→C4 measures the bridge, and C1→C4 is descriptive only.

Our contributions are:

- evidence that fine-tuning raises accuracy from 48.5% to 82.5%;
- a matched bridge comparison showing a fall to 49.5%, with 12 corrections but 78 harmful overturns;
- trace-supported evidence that translation failures and semantically divergent executable logic explain the degradation; and
- a provenance-checked analysis that regenerates tables, intervals, and figures from canonical predictions.

II. Related Work

A. Logical reasoning benchmarks

ProofWriter operationalizes open-world deductive classification, records proof depth, and defines Unknown when neither a conclusion nor its contradiction is derivable [1]. FOLIO provides naturally written arguments paired with first-order-logic annotations [5]; it is not evaluated in this study.

B. Language models and symbolic solvers

Chain-of-thought prompting exposes intermediate natural-language reasoning steps [6]. Logic-LM translates problems into symbolic forms and can use solver feedback for refinement [2]; LINC delegates deduction to a theorem prover after semantic parsing [3]. NL2Logic uses AST-guided recursive translation and deterministic two-pass compilation [4]. We evaluate the latter translation pattern without comparing unmatched scores from prior systems.

C. Efficient adaptation

Qwen3 includes a 1.7B-parameter dense model [7]. LoRA learns low-rank updates while pretrained weights remain frozen [8]; QLoRA combines such adapters with 4-bit NF4 quantization and paged optimization [9]. We use these mechanisms to keep base-model identity fixed across conditions.

III. System and Implemented Conditions

A. Task formulation

ProofWriter uses open-world three-way classification [1]. For a theory T and hypothesis h, the target label is

$$y(T,h) = \begin{cases} \text{True}, & T \models h, \\ \text{False}, & T \models \neg h, \\ \text{Unknown}, & \text{otherwise}. \end{cases} \tag{1}$$

All headline metrics use the three gold classes. A malformed or missing prediction is assigned to an Invalid prediction column and cannot match any gold label.

B. C1: base direct

C1 loads the pinned Qwen3-1.7B revision with NF4 quantization [7], [9] and deterministic decoding. An answer-only prompt requests True, False, or Unknown; no adapter or symbolic component is used.

TABLE I
MODEL AND FINE-TUNING CONFIGURATION.

Item	Value
Base model	Qwen3-1.7B
Quantization	NF4, FP16 compute
Training examples	2,000
Validation examples	2,000
LoRA rank / alpha / dropout	16 / 32 / 0.05
Learning rate / epochs	2.0e-04 / 3
Batch / accumulation	1 / 16
Optimizer / schedule	paged AdamW 8-bit / cosine
Selected adapter	checkpoint-250
Bridge	descriptive NL2Logic → AST → Z3

TABLE II
DATASET ACCOUNTING. FOLIO IS PREPARED BUT NOT EVALUATED.

Dataset	Train	Validation	Held-out
ProofWriter source	585,552	85,468	174,476
ProofWriter selected	2,000	2,000	2,000
ProofWriter canonical	–	–	200
FOLIO prepared	896	105	203

C. C3: fine-tuned direct

C3 loads the same base revision and a ProofWriter LoRA adapter from checkpoint-250 [1], [8]. Training uses NF4 quantization and paged optimization following the QLoRA design pattern [9]. C3 uses the same direct prompt and evaluation path as C1.

D. C4: fine-tuned neuro-symbolic bridge

C4 loads the exact checkpoint-250 adapter used by C3. The canonical metadata identifies the descriptive prompt/schema configuration. Here, descriptive bridge means only this fixed natural-language schema variant; thereafter we use symbolic bridge. Following NL2Logic, it recursively constructs typed FOL AST nodes and compiles them in two passes [4], then queries the Z3 SMT solver [10]. Entailment, contradiction, or neither maps to True, False, or Unknown, and that symbolic verdict is final.

IV. EXPERIMENTAL SETUP

A. Data and canonical evidence

The prepared ProofWriter source [1] contains 585,552 training, 85,468 validation, and 174,476 test records. Controlled-language manifests select 2,000 training examples, 2,000 training-validation examples, and a 2,000-example held-out pool. The canonical evaluation is a label-and-QDep-stratified sample of 200 held-out examples using seed 42: 53 True, 53 False, and 94 Unknown.

The sole experimental source is the canonical ProofWriter run [1] at `results/final200_proofwriter`. Each condition contains 200 predictions with identical ordered identifiers, gold labels, and frozen subset hash. Provenance checks also require C3 and C4 to reference the same checkpoint-250 path; older and incomplete runs are excluded.

B. Metrics and statistical testing

We report strict accuracy, macro precision/recall/F1, weighted F1, classwise scores, valid-output rate, and confusion matrices. Valid-only accuracy is reported only as a diagnostic and never replaces strict accuracy. For condition c over $n = 200$ examples,

$$\mathrm{Acc}_c = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\hat{y}_{c,i} = y_i]. \tag{2}$$

Confidence intervals use 10,000 paired bootstrap resamples with seed 42. The C3/C4 correctness difference is tested using the exact two-sided McNemar/binomial test on discordant pairs. The test concerns final correctness, not the internal cause of each error.

For intervention analysis we partition pairs into wrong→correct (W→C), correct→correct, correct→wrong (C→W), and wrong→wrong. The correction rate is W→C divided by all C3 errors; the harm rate is C→W divided by all C3 correct answers.

V. RESULTS

A. Fine-tuning effect: C1 to C3

Table III and Fig. 2 show a large fine-tuning gain. Accuracy increases from 48.5% (97/200) to 82.5% (165/200), an absolute improvement of 34.0 percentage points. Macro-F1 increases from 0.251 to 0.826. The base model strongly defaults to Unknown: it achieves Unknown F1 of 0.646 but True F1 of 0.107 and False F1 of 0.000. Fine-tuning produces balanced classwise F1 scores of 0.854, 0.804, and 0.821 for True, False, and Unknown, respectively.

B. Bridge effect: C3 to C4

Routing the same checkpoint through the bridge reduces strict accuracy from 82.5% to 49.5%, a 33.0-point decrease. Macro-F1 falls by 0.367, from 0.826 to 0.459. The paired bootstrap 95% confidence interval for the accuracy difference is $[-0.410, -0.245]$, and exact McNemar testing on 90 discordant pairs gives $p = 5.20 \times 10^{-13}$. The direct model makes 35 errors; C4 makes 101, corresponding to a 188.6% increase in the error rate. C1 and C4 differ by only one accuracy point, but that contrast is descriptive because both adaptation and inference path differ.

C4 produces a valid label for 179 examples (89.5%). Conditional accuracy on those valid outputs is 55.3%, still far below C3. Thus, invalid outputs explain 21 errors but not the majority of the 66 additional errors.

C. Confusion structure and reasoning depth

Figure 3 shows that C3 resolves True and False with few cross-class mistakes, while C4 shifts many cases toward Unknown: 29 of 53 True and 34 of 53 False instances become Unknown. C4 also maps 14 Unknown examples to True and yields 21 invalid predictions. This pattern is consistent with information being lost or distorted before symbolic inference.

QDep metadata support an observational depth analysis. C3 leads at every grouped depth: it moves from 96.2% at depth 0

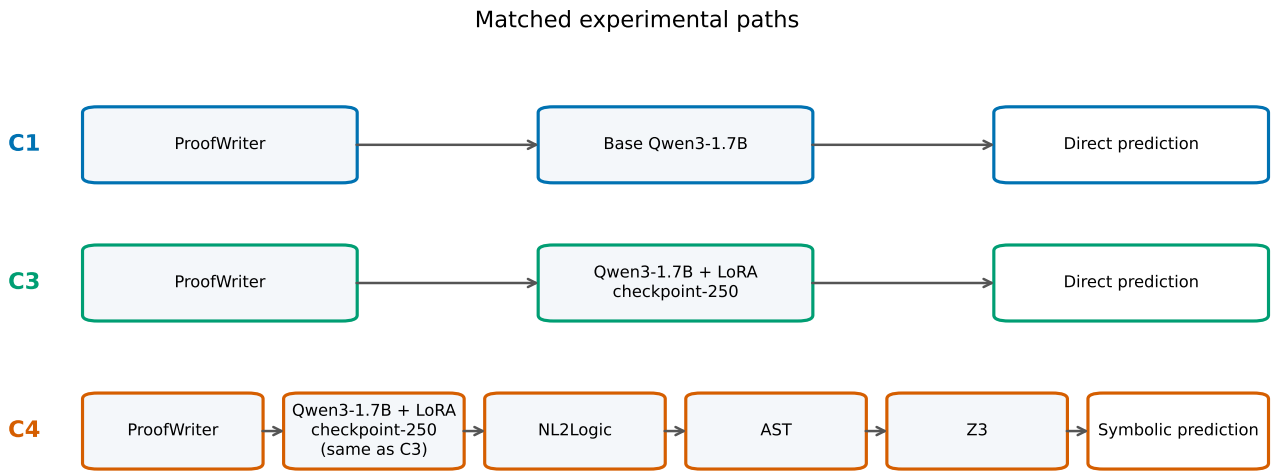


Fig. 1. Matched experimental paths. C3 and C4 use the same checkpoint-250 LoRA adapter [8]; only C4 adds the symbolic translation and solver path.

TABLE III
STRICT CANONICAL PROOFWRITER RESULTS. INVALID OUTPUTS COUNT AS ERRORS. ACCURACY AND VALIDITY ARE PERCENTAGES.

Condition	Correct	Accuracy	Macro-F1	Weighted F1	Valid
C1: base direct	97/200	48.5	0.251	0.332	100.0
C3: fine-tuned direct	165/200	**82.5**	**0.826**	0.825	100.0
C4: fine-tuned bridge	99/200	49.5	0.459	0.491	89.5

TABLE IV
PER-CLASS STRICT PERFORMANCE.

Cond.	Class	Prec.	Rec.	F1	Support
C1	True	1.000	0.057	0.107	53
C1	False	0.000	0.000	0.000	53
C1	Unknown	0.477	1.000	0.646	94
C3	True	0.880	0.830	0.854	53
C3	False	0.837	0.774	0.804	53
C3	Unknown	0.792	0.851	0.821	94
C4	True	0.500	0.321	0.391	53
C4	False	1.000	0.226	0.369	53
C4	Unknown	0.526	0.745	0.617	94

TABLE V
BOOTSTRAP INTERVALS (10,000 PAIRED RESAMPLES).

Quantity	Estimate	95% CI
C1 accuracy	0.485	[0.415, 0.555]
C1 macro-F1	0.251	[0.208, 0.299]
C3 accuracy	0.825	[0.770, 0.875]
C3 macro-F1	0.826	[0.769, 0.876]
C4 accuracy	0.495	[0.430, 0.565]
C4 macro-F1	0.459	[0.380, 0.534]
C4−C3 accuracy	-0.330	[-0.410, -0.245]
McNemar discordant	90	$p = 5.20e-13$

to 26.7% at depth 3 and 75.0% in the four-example 4+ group. C4 moves from 69.2% at depth 0 to 6.7% at depth 3 and 0% at 4+. The imbalanced and very small tail groups preclude causal conclusions.

TABLE VI
PAIRED C3→C4 INTERVENTION OUTCOMES.

Outcome	Count
Wrong → correct	12
Correct → correct	87
Correct → wrong	78
Wrong → wrong	23
Correction rate	34.3%
Harm rate	47.3%
Net corrections	-66

VI. INTERVENTION AND ERROR ANALYSIS

A. *Corrections versus harmful overturns*

The paired transition counts in Table VI reveal why aggregate performance collapses. C4 corrects 12 of C3's 35 errors, a 34.3% correction rate. However, it overturns 78 of C3's 165 correct answers, a 47.3% harm rate. Eighty-seven examples remain correct and 23 remain wrong, yielding 66 net lost corrections. A useful bridge policy would need to identify when translation and symbolic execution are trustworthy; no such policy exists in this implementation.

B. *Bridge reliability and failure modes*

Twenty C4 examples fail before being run by a solver: 11 are malformed and nine are schema invalid. Of the remaining

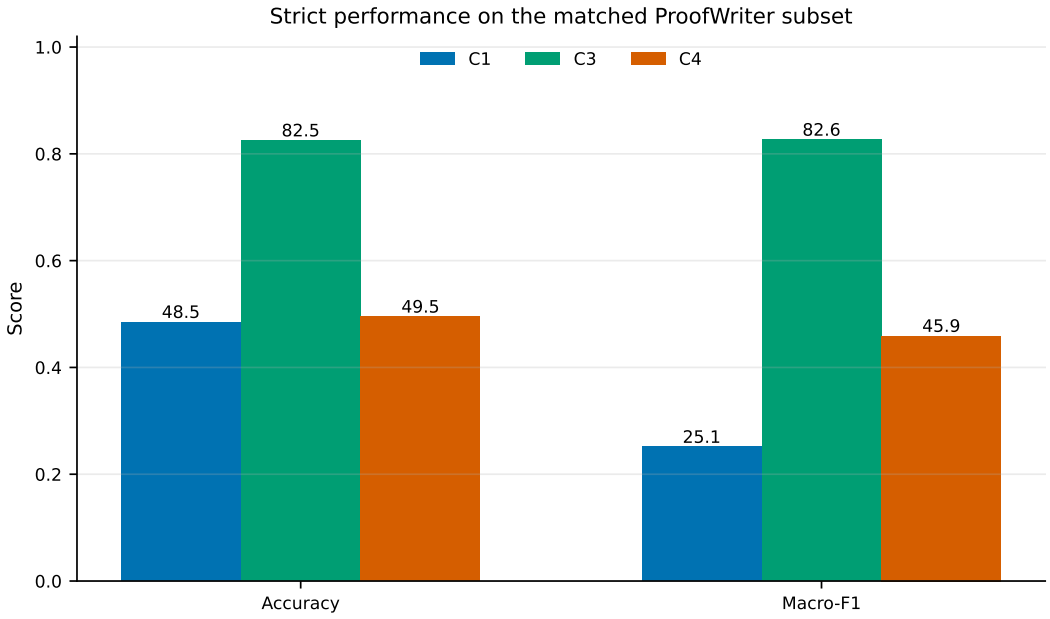


Fig. 2. Strict accuracy and macro-F1. C1→C3 isolates fine-tuning; C3→C4 isolates the bridge.

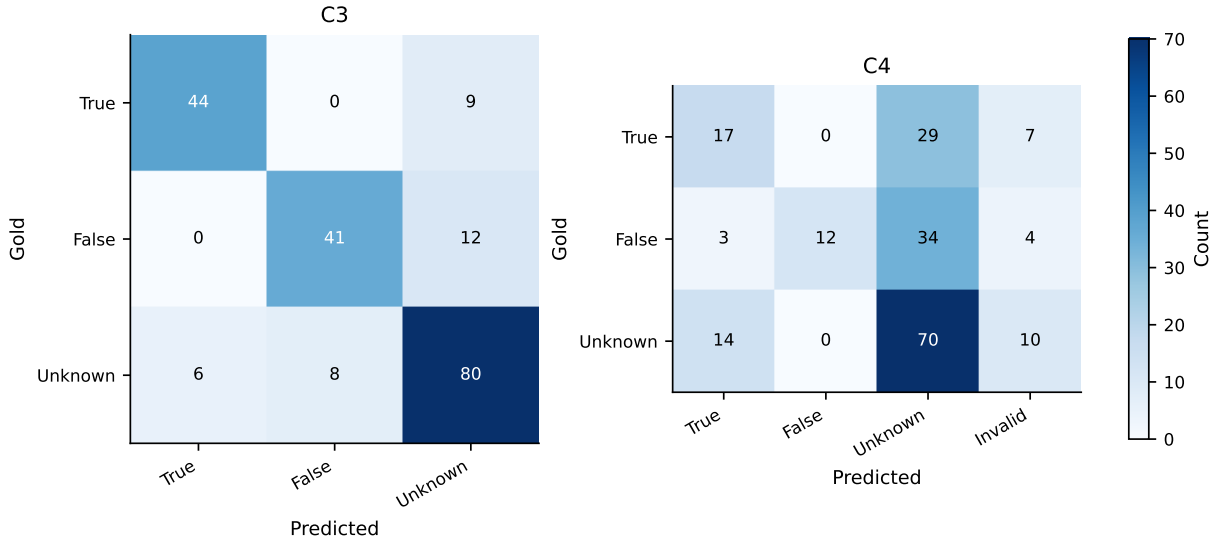


Fig. 3. Strict confusion matrices for C3 and C4. Invalid is a prediction outcome, not a fourth gold class.

180, 179 execute successfully in Z3 and one returns inconsistent premises. Yet, 80 of the 179 executable examples (44.7%) return an incorrect symbolic conclusion.

Translation diagnostics yield direct evidence of mismatch. The recorded semantic equivalence field is `different` for 158 examples and unavailable for 42; exact and normalized logic matches are both zero. Parser recovery fields appear in 1,122 steps of trace data for 193 examples. These are step-level recovery events (not 1,122 failed examples), but the prevalence of permissive fallback behavior in the translation path is evident. We do not attribute every executable error to a particular parser step: solver-correct reasoning over semantically altered logic, hypothesis mistranslation, rule mistranslation, and predicate-argument inconsistency can yield the same ultimate symptom.

C. Latency

Direct inference is inexpensive: mean end-to-end latency is 0.136 seconds for C1 and 0.228 seconds for C3. C4 averages 22.60 seconds (median 21.27, 95th percentile 38.71), about 99 times C3 latency. Across the 200 examples, C4 consumes roughly 75.3 minutes and records 11,726 recursive parser call requests and 3,960 model invocations. The current

TABLE VII
C4 BRIDGE RELIABILITY AND OUTCOMES ($n = 200$).

Stage/outcome	Count	Rate (%)
Schema/parser/AST valid	180	90.0
Compilation success	180	90.0
Z3 executable	179	89.5
Correct symbolic verdict	99	49.5
Executable but incorrect	80	40.0
Malformed/schema invalid	20	10.0
Inconsistent premises	1	0.5

bridge therefore trades both accuracy and latency for symbolic executability.

VII. Discussion

The C1/C3 comparison shows that it is the 2,000-example LoRA adaptation that is the dominant positive contributor in this experiment: it eliminates C1's strong Unknown bias in favor of balanced True/False/Unknown performance. The matched C3/C4 comparison then pinpoints the added translation-and-solver path as the source of the degradation, not a differing checkpoint.

The bridge is not uniformly detrimental—it fixes 12 errors—but its 78 harmful overturns suggest that the current translator is unable to reliably determine when its formalization has preserved the source theory. Future work should consider semantic consistency, predicate/variable binding, and calibrated abstention as novel matched conditions. This conclusion is specific to the implemented bridging mechanism and should not be construed as a general statement about the negative impact of neuro-symbolic components on reasoning.

VIII. Limitations and Threats to Validity

The canonical subset contains 200 examples from one controlled benchmark and one random seed. Although matched design supports paired inference, confidence intervals remain wide for small depth groups. ProofWriter's synthetic language limits generalization to naturally occurring text. The base model, adapter, prompts, and bridge are single implementations; results do not establish that fine-tuning always dominates neuro-symbolic reasoning or that NL2Logic generally fails.

The trace fields can establish well-formedness, execution, fallback activity, and recorded semantic mismatch, but they cannot uniquely assign causal responsibility for every incorrect verdict. Moreover, C4 has no retain/correct/abstain policy, so this study evaluates complete routing through the bridge rather than selective verification.

An additional natural-language benchmark is prepared for future work, but no matched prediction-level evaluation exists; no performance is reported. No matched self-check or symbolic-only baseline is available.

IX. Conclusion

In a matched evaluation using 200 examples of ProofWriter, the accuracy of Qwen3-1.7B is increased from 48.5% to 82.5% by means of LoRA fine-tuning. When the same checkpoint is passed through the symbolic bridge that has been implemented, accuracy drops to 49.5%. The bridge rectifies 12 direct errors but causes 78 answers which were originally correct to become incorrect; parser failures account for 20 of the invalid cases, while many of the executable verdicts are still wrong. The conclusion drawn is that the bottleneck in this implementation is not symbolic reasoning in general but translation fidelity.

Future Work

Future work should be directed at enhancing the semantic fidelity during the NL2Logic translation phase before carrying out symbolic inference. Possible approaches might involve constrained decoding, explicit checks on predicates and variable bindings, stronger validation of the generated logical forms, and calibrated abstention when translation confidence is low. It could also be investigated to apply a selective verification policy so that symbolic reasoning is used only when the formalisation is sufficiently reliable, rather than always substituting the direct prediction from the model. The matched evaluation should also be extended to include FOLIO and other natural-language reasoning benchmarks to determine whether the failure modes observed with ProofWriter generalise beyond that system.

References

[1] O. Tafjord, B. D. Mishra, and P. Clark, "ProofWriter: Generating implications, proofs, and abductive statements over natural language," *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 2021. [Online]. Available: https://arxiv.org/abs/2012.13048

[2] L. Pan, A. Albalak, X. Wang, and W. Y. Wang, "Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning," *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. [Online]. Available: https://arxiv.org/abs/2305.12295

[3] T. X. Olausson, A. Gu, B. Lipkin, C. E. Zhang, A. Solar-Lezama, J. B. Tenenbaum, and R. Levy, "LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers," *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. [Online]. Available: https://arxiv.org/abs/2310.15164

[4] R. R. Putra, R. S. P. Basuki, Y. Cheng, and P. Gao, "NL2Logic: AST-guided translation of natural language into first-order logic with large language models," in *Findings of the Association for Computational Linguistics: EACL 2026*, 2026, pp. 6035–6051.

[5] S. Han, H. Schoelkopf, Y. Zhao, Z. Qi *et al.*, "FOLIO: Natural language reasoning with first-order logic," *arXiv preprint arXiv:2209.00840*, 2022.

[6] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, vol. 35, 2022. [Online]. Available: https://arxiv.org/abs/2201.11903

[7] Qwen Team, "Qwen3 technical report," *arXiv preprint arXiv:2505.09388*, 2025.

[8] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.

[9] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: https://arxiv.org/abs/2305.14314

[10] L. de Moura and N. Bjorner, "Z3: An efficient SMT solver," in *Tools and Algorithms for the Construction and Analysis of Systems*, ser. Lecture Notes in Computer Science, vol. 4963. Springer, 2008, pp. 337–340.